SAI KISHORE
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SUMMARY

Hands-on engineer with a background in electrical and embedded systems, experienced in testing, diagnostic analysis, and data acquisition. Skilled in handling real-time sensor data, system tracing, and performance monitoring. Worked with embedded Linux, telemetry tools, and platform-level debugging. Proven ability in root cause analysis, durability data handling, and integrating diagnostic modules into backend platforms. Proficient in coordinating testing teams, analyzing technical issues, and ensuring performance reliability in simulated and live environments.

PROFESSIONAL EXPERIENCES

UHCL | Houston, Texas
Research Assistant | September 2023 - December 2024
Worked on backend diagnostic modules and embedded Linux-based system services. Participated in system tracing, log analysis, and service behavior simulation. Developed real-time monitoring dashboards using OpenTelemetry and ELK Stack for identifying and analyzing runtime anomalies. Assisted in integration testing and platform-level fault analysis under simulated data conditions.

Infosys Limited | Mysore, India
Systems Engineer | November 2022 - June 2023
Collaborated on backend systems interfacing with simulated sensor data and distributed control modules. Supported cloud-hosted microservices related to system health checks, performance metrics, and diagnostics. Engaged in structured debugging, root cause identification, and log review. Worked on service flow tracing and cross-module signal communication in modular architectures.

Wipro Limited | Bengaluru, India
Project Engineer | January 2022 - June 2022
Participated in development of full-stack systems integrated with hardware interaction modules. Focused on signal processing, logging, and system alerts through embedded-like services. Aided in backend support of telemetry-style APIs and diagnostic services. Worked with cross-functional teams to validate and test backend responses under simulation-like conditions.

RESPONSIBILITIES

- Conducted root cause analysis through the collection and analysis of structured backend logs, along with simulated sensor data. Employed performance traces and system outputs to detect fault sites in service interactions and logged anomalies within testing intervals.
- Developed diagnostics and telemetry backend modules to facilitate real-time testing of the system. Added log fetching, data analysis, and fault-tracking features to allow continuous evaluation of system performance under simulated vehicular conditions.
- Developed embedded Linux environments and containerized systems to simulate RTOS-like responses under load testing. Concentrated on developing modular services that simulated control systems under different road simulation inputs.
- Combined ELK Stack and OpenTelemetry for building a real-time system monitoring solution for system health, diagnostics, and signal analysis. Used dynamic dashboards and alerting mechanisms for tracking sensor deviations and runtime exceptions in test simulations.
- Conducted simulation of processing involved in acceleration, displacement, and load-pressure data in test conditions. Used Python scripts and backend logic to simulate output behavior of transducers while maintaining proper signal transfer between components.
- Designed and developed APIs and microservices for collecting and storing test data from simulated components. Maintained high fidelity of data through continuous signal sampling, error handling, and response monitoring in simulated real-world road test conditions.
- Engaged back-end testing procedures that involved client-server paradigm communication models for telemetry-type data exchange. Used asynchronous messaging methods to mimic real-time processing behaviors of vehicle signals.
- Aided in the creation of modular diagnostic utilities that interfaced with test scripts and container environments. Provided logging correctness and system trace coordination for effective bug detection while conducting durability testing.
- Used containerized environments to simulate vehicle-level system integration. Was interested in consistency of performance across backend services in processing input from simulated transducer-based measurements.
- Worked with databases to hold vehicle telemetry-like data such as pressure, displacement, and vibration readings. Designed schema for optimal query performance in repeated signal analysis and log review.

- Aided in backend component testing under load profiles based on chassis and suspension behaviors. Replicated road conditions using different signal inputs and recorded backend stability during test sequences.
- Built data pipelines that duplicated the acquisition systems for road load data. Checked the integrity of streaming input through thorough testing and used cleanup scripts to ensure test data accuracy and filtration.
- Engaged in integration testing simulating the embedded diagnostics use cases. Tested logging and tracing modules within controlled environments to identify bottlenecks, faults, and inconsistencies between components.
- Participated in internal validation discussions and test result evaluation. Prepared detailed summaries on data inconsistencies, supported test engineers through insights, and documented test results for team presentation.
- Collaborated with cross-functional test teams of software engineers and QA testers. Coordinated test environment setup, result interpretation, and post-test diagnostic report generation to improve overall test reliability.

EDUCATION

University of Houston Clear-Lake, Houston, Texas **| Dec2024**
Masters in Computer Engineering **| GPA: 3.6**
Lendi College of Engineering & Technology, Andhra Pradesh
Bachelors in Electrical, Electronics Engineering **| GPA: 3.2**

TECHNICAL SKILLS

Programming & Data Analysis: Python, C, C++, Java, MATLAB, Simulink, Shell Scripting, SQL, Pandas, NumPy, Data Visualization, Signal Processing, Real-Time Data Monitoring.
Vehicle Systems & Test Engineering: Road Load Data Acquisition, Vehicle Testing, Durability Testing, Transducer Data Processing, Chassis/Suspension/Driveline System Analysis, Root Cause Analysis, Data Reduction, Diagnostic Tools, MTS Road Simulator Exposure.
Tools & Platforms: OpenTelemetry, ELK Stack (Elasticsearch, Logstash, Kibana), Prometheus, Grafana, Docker, Git, Jenkins, AWS, VS Code, CAN Protocol, Sensor Integration (Accelerometers, Load Cells, Pressure Sensors), Emulator-Based Testing.
Database & Backend Systems: PostgreSQL, MySQL, MongoDB, RESTful APIs, Backend Telemetry Pipelines, System Logging, Test Signal Storage Architecture, Real-Time Fault Tracking, API Debugging.

CERTIFICATIONS

Advanced JavaFull Stack - StackRoute
Internet of Things (IoT) - Origin, ITsez Hub
Automate the Boring Stuff with Python - UDEMY
Mathematical Methods and its Applications - NPTEL
Microsoft Azure Solutions Architect Expert - LinkedIn
Programming with Python - Internshala
JavaScript Certification - UDEMY
Switching Theory and Logic Design - NPTEL

PROJECTS

Vehicle Signal Monitoring & Diagnostic Platform: Designed a test backend to simulate and process real-time data from vehicle sensors. Simulated CAN-like messages and used Python scripts to parse acceleration, load, and displacement signals. Integrated OpenTelemetry and ELK Stack for live visualization and diagnostics. Validated data fidelity and created an alerting system for runtime fault events.
Skin Cancer Detection (Edge AI + Sensor Simulation): Built a deep learning model for detecting cancerous skin lesions. Extended the system for edge-device simulation by mimicking sensor input (camera feed) and real-time classification output. Simulated signal-based inference timing and tested model accuracy performance constraints.
Durability Test Automation with Embedded Modules: Developed backend modules that mimicked real-time durability test environments. Used Docker containers and emulated sensor input streams to test multi-point displacement and stress conditions. Integrated logs into Grafana dashboards and generated reports for anomaly events under test loads.
End-to-End Full-Stack Monitoring Application: Built a full-stack app using React, Node.js, and MongoDB to monitor signal data. Integrated REST APIs to simulate data ingestion from virtual vehicle components. Deployed the app on AWS with CI/CD pipelines and real-time UI updates for diagnostic values and alerts.

HONORS AND CONTRIBUTIONS

University of Houston-Clear Lake, Houston, Texas
Platform Software & Embedded Systems Research | Sep 2023 – Dec 2024

- Participated in research projects focusing on test signal acquisition and diagnostic system development in embedded Linux environments.
    - Developed a modular data monitoring framework using OpenTelemetry and ELK Stack for real-time performance visibility and system fault detection.
- Published findings in IEEE conference proceedings focused on anomaly detection, fault analysis, and diagnostic automation for high-performance systems.
- Successfully identified and reported a critical system bug in UHCL’s infrastructure platform, received special recognition and was promoted to Tech Administrator.